Causal Inference Crash Course:

Part 6 — Panel Models: DiD, Synthetic Control & Synthetic DiD

Julian Hsu

Causal Inference Series

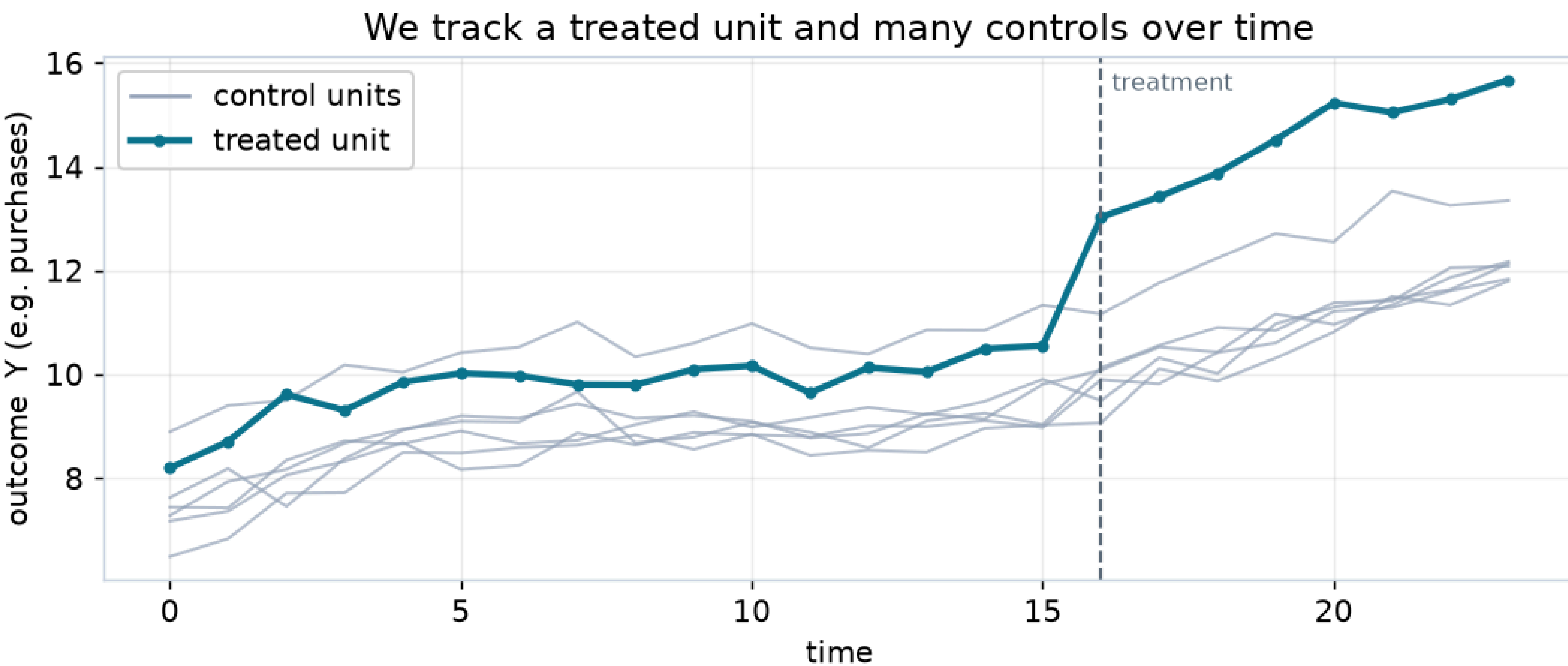
1. Foundations
2. Causal Models
3. Inference
4. Best Practices
5. Heterogeneous Treatment Effects
6. **Panel Models ← this deck**
7. Regression Discontinuity
8. Surrogate Models
9. Arguable Validation
10. Conformal Inference

Overview

- Panel methods use **repeated observations of the same units over time** to estimate causal effects — the workhorse of quasi-experimental design.
  - A quarter of NBER working papers and 16% of top-five journal articles use difference-in-differences (Currie, Kleven, Zwiers 2020).
- We cover three ways to build the **counterfactual** for a treated unit:
  - **Difference-in-Differences (DiD)** — a parallel control trend
  - **Synthetic Control (SC)** — a weighted average of controls
  - **Synthetic DiD (SDID)** — a hedge that borrows from both
- Throughout: define the **estimand first**, state the assumptions plainly, then estimate — and stress-test everything in a simulated worked example.
- Assumes the earlier decks: potential outcomes, OLS, inference.

Panel Data

The big picture



- We track a **treated** unit and many **control** units over time, and see their outcomes before and after treatment.

- If, absent treatment, the treated unit's trend would have matched the controls, then the **gap that opens after treatment is the effect**.

The target parameter: the ATT

- Each unit has two potential outcomes at time t: $Y_{it}(1)$ if treated, $Y_{it}(0)$ if not. We only ever see one.
- The problem is the **missing** $Y_{it}(0)$ for treated units after treatment.
- Panel methods target the **Average Treatment effect on the Treated**:

$$\tau_{ATT} = E\Big[\,Y_{it}(1) - Y_{it}(0) \;\Big|\; W_i = 1,\ t \geq 0\,\Big]$$

- This is **not** the ATE. We learn the effect *for the units that were treated*, by reconstructing their counterfactual $Y_{it}(0)$.
- Every model below is just a different way to **predict that counterfactual**.

Comparing trends lets us *arguably* validate

- We want to know how the treated unit would behave **if we had not treated it**.
- We can never check this directly — but we *can* check the **pre-treatment** period, where the counterfactual is observed.
- The more the control trend matches the treated trend **before** treatment, the more we trust it to stand in for $Y_{it}(0)$ **after**.
- This is the recurring move in every panel method: earn trust in the pre-period, then extrapolate.

Where DiD, SC, and SDID come in

- How do we build that ideal control trend — and what if no single control unit looks like the treated one?

Method	Counterfactual is…	Trusts…
DiD	the control group shifted to the treated unit's level	parallel trends
Synthetic Control	a **weighted** average of controls matching the pre-period	pre-period fit
Synthetic DiD	a weighted control **plus** a DiD-style level shift	a bit of both

- We build them in that order, because each one relaxes a weakness of the last.

Difference-in-Differences

DiD: we predict the counterfactual

- The treated unit's missing $Y_{it}(0)$ is predicted by taking its **own pre-treatment level** and adding the **change the control group experienced**.
- In words: *"the treated units would have moved just like the controls did."*

Setup and notation

- Let's set up notation:
  - $W_i \in \{0, 1\}$ — whether unit i is **ever** treated.
  - $\text{post}_t = 1\{t \geq 0\}$ — whether period t is after treatment.
  - $D_{it} = W_i \cdot \text{post}_t$ — the **treatment indicator** (on only for treated units, in post periods).
- The data-generating model we have in mind:

$$Y_{it} = \tau D_{it} + \gamma_t + \eta_i + \zeta_{it}$$

  - η_i: a **unit** fixed effect (time-invariant level of unit i)
  - γ_t: a **time** fixed effect (common shock in period t)
  - ζ_{it}: everything else — the idiosyncratic error

Two naïve comparisons, and why each fails

- **Post vs. Pre** (treated units only): compare Y_{it} after vs. before.
    - Fails: γ_t moved too — you can't tell the treatment from the trend.
- **Treated vs. Control** (post only): compare treated to control after treatment.
    - Fails: η_i differs — treated and control units start at different levels.
- Each naïve comparison confounds the treatment with *one* nuisance term. The trick is to **difference both away**.

DiD comes from combining two differences

$$Y_{it} = \tau D_{it} + \gamma_t + \eta_i + \zeta_{it}$$

- **Post – Pre**, separately within the treated and the control groups:

$$\Delta_T = E[Y_{it} \mid t \geq 0, W_i{=}1] - E[Y_{it} \mid t < 0, W_i{=}1]$$

$$\Delta_C = E[Y_{it} \mid t \geq 0, W_i{=}0] - E[Y_{it} \mid t < 0, W_i{=}0]$$

- Differencing over time **cancels the unit effect** η_i, leaving only the time term and the error:

$$\Delta_T = \tau + (\gamma_{t\geq 0} - \gamma_{t<0}) + \text{errors}, \qquad \Delta_C = (\gamma_{t\geq 0} - \gamma_{t<0}) + \text{errors}$$

Deriving DiD: the second difference

$$\Delta_T = \tau + (\gamma_{t\geq 0} - \gamma_{t<0}) + \bar{\zeta}_T, \qquad \Delta_C = (\gamma_{t\geq 0} - \gamma_{t<0}) + \bar{\zeta}_C$$

- The **time term γ_t is common** to both groups — so subtract them:

$$\Delta_T - \Delta_C = \tau + (\bar{\zeta}_T - \bar{\zeta}_C)$$

- If treated and control units do **not** differ in time-varying ways ($E[\bar{\zeta}_T - \bar{\zeta}_C] = 0$), the second difference **is** the ATT:

$$\boxed{\Delta_T - \Delta_C = \tau_{ATT}}$$

Estimating DiD: two-way fixed effects

- A simple version aggregates the fixed effects into group/period dummies:

$$Y_{it} = \tau D_{it} + \gamma\, \mathbf{post}_t + \eta\, W_i + \epsilon_{it}$$

- The current standard is the **two-way fixed effects (TWFE)** regression, with a full set of unit and time fixed effects:

$$Y_{it} = \tau D_{it} + \gamma_t + \eta_i + \epsilon_{it}$$

- The coefficient $\hat{\tau}$ on D_{it} **is** the diff-in-diff estimate — same number as the two differences, with more precision and easy covariates.
- In `panelib`: `did.twfe(...)`.

The assumptions, stated plainly

- DiD identifies the ATT under **two** assumptions — keep them separate:
- **1. Parallel trends.** Absent treatment, the treated group's average outcome would have moved *parallel* to the control group's:

$$E[Y_{it}(0) \mid W_i{=}1] - E[Y_{it}(0) \mid W_i{=}0] \textbf{ is constant in } t.$$

- **2. No anticipation.** Units don't change behavior *before* treatment starts (no effect leaking into the pre-period), so the pre-period identifies $Y(0)$.
- (And, as everywhere: **no spillovers** — treatment must not touch the control units' outcomes, or their trend is no longer $Y(0)$. Think geo launches leaking across borders.)
- A subtlety: parallel trends is **functional-form dependent** — it can hold in levels but not in logs. Pick the scale your estimand is about (Roth & Sant'Anna 2023).

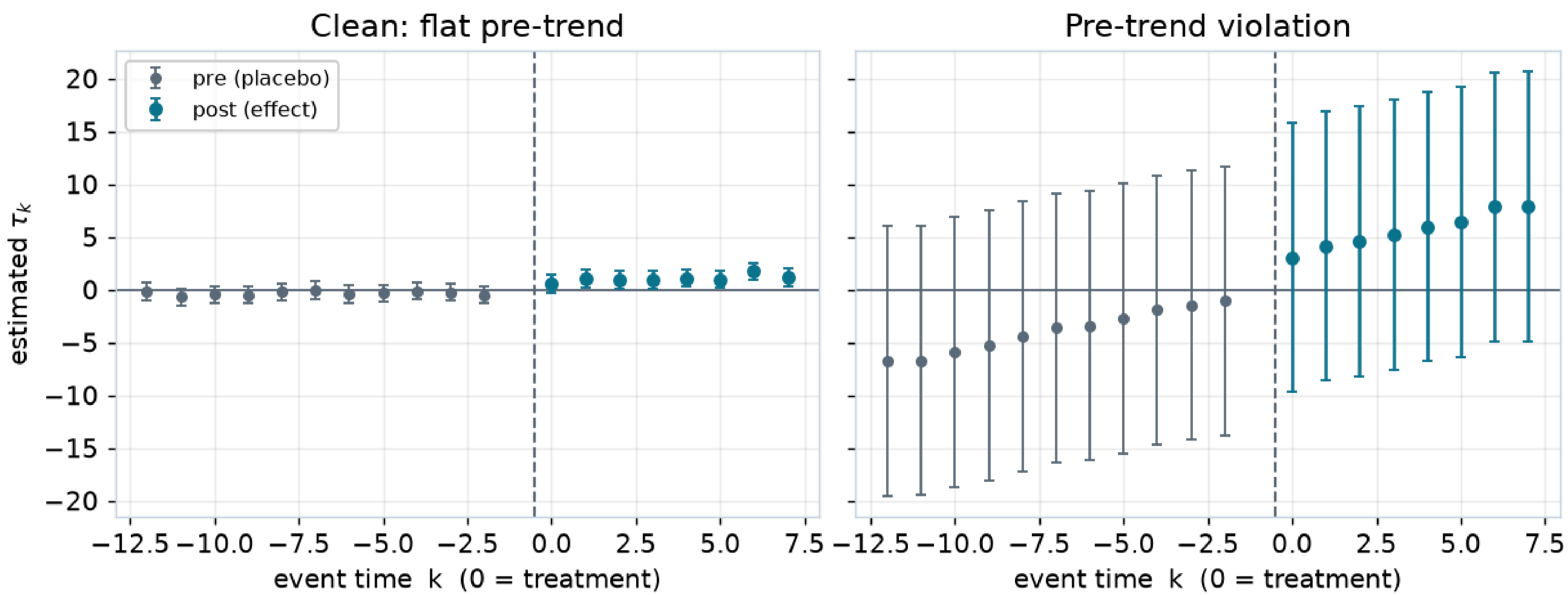
DiD estimates the ATT, not the ATE

- The second difference recovers τ **only for the treated units** — their realized trend is what we differenced.
- Nothing here tells us the effect on a control unit that was never treated; their potential response could differ.
- This is a *feature*, not a bug: the ATT is usually the policy-relevant number ("what did the rollout do to the units we rolled it out to?").

The event study: check trends period-by-period

- Instead of one pre and one post, estimate a **separate coefficient for each period** relative to treatment ($k = 0$):

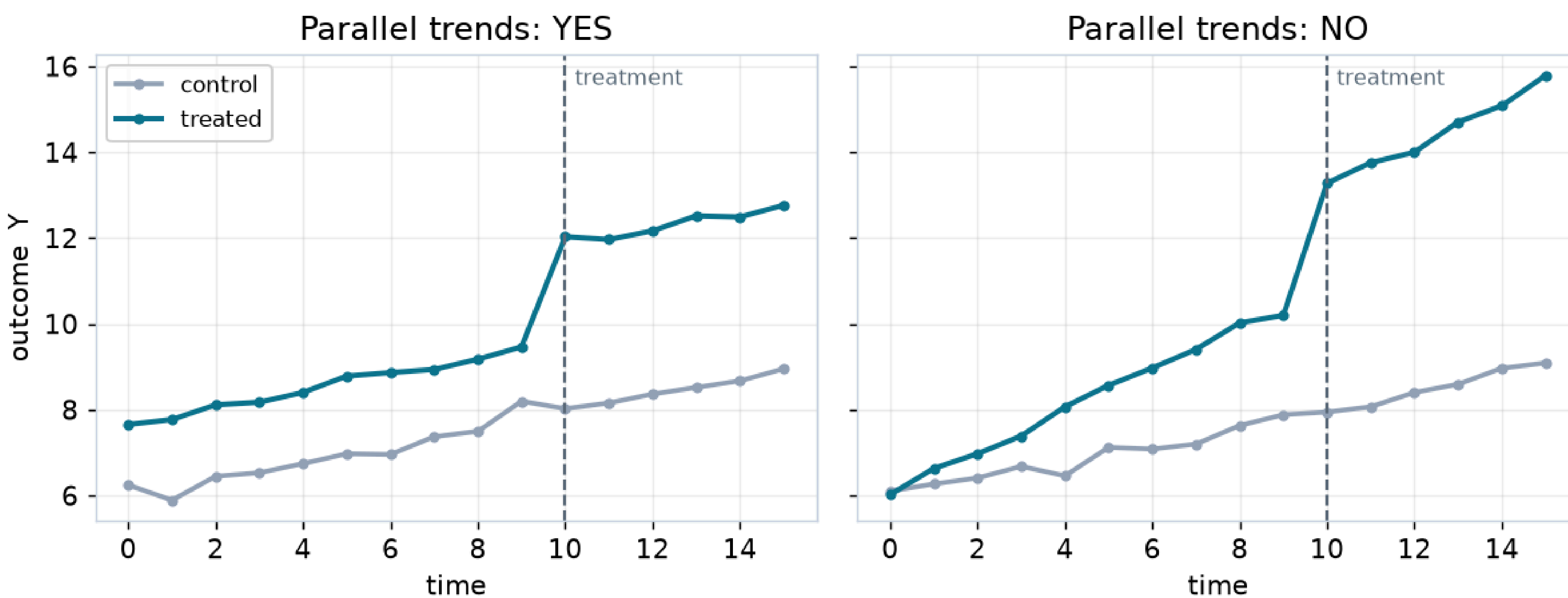
$$Y_{it} = \sum_{k \neq -1} \tau_k 1\{t - t_i^* = k\} + \gamma_t + \eta_i + \epsilon_{it}$$



- **Pre-period τ_k ($k < 0$) should sit at zero** — that is the parallel-trends check. **Post-period τ_k** traces the dynamic effect.

Validating DiD: is the parallel-trends assumption plausible?

- **1. Eyeball it.** Plot treated vs. control; do the pre-treatment trends move together? (Figure below, left = yes, right = no.)
- **2. Placebo pre-trend test.** In the event study, jointly test that the pre-period coefficients are zero.



A caution on pre-trend tests

- A passing pre-trend test is **reassuring, not proof** — the assumption is about the *unobserved* post-period counterfactual, which no test can see.
- **Pre-tests have low power** (Roth 2022): the trend violations most likely to bias you are exactly the ones a pre-test is least likely to reject. And conditioning on "passing" can itself distort your estimate.
- Best practice: report the event study, and probe **sensitivity** to plausible violations rather than treating a non-rejection as a green light (Rambachan & Roth 2023).

Including time-varying covariates

- Parallel trends may hold only **after conditioning** on covariates X_{it} (e.g. seasonality, unit size).
- Add them to the TWFE regression — but a plain linear control can be biased if effects are heterogeneous.
- Modern practice uses **doubly-robust DiD** (Sant'Anna & Zhao 2020): combine an outcome-regression counterfactual with a propensity model, so you only need *one* of them right.
- Caution: condition only on covariates **treatment cannot touch** (pre-treatment values are safest). Controlling for a post-treatment variable absorbs part of the effect itself — the classic *bad control*.

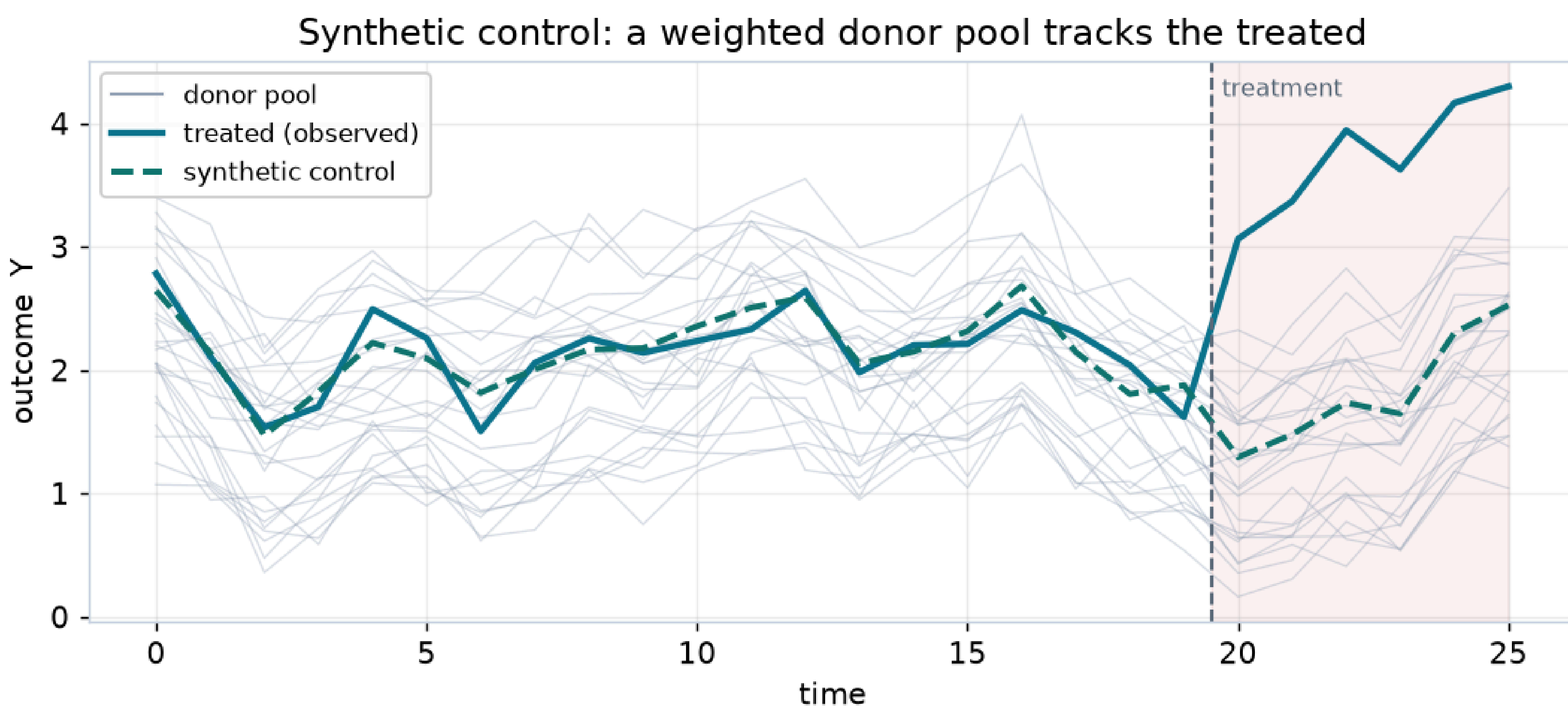
⚠️ Staggered adoption breaks plain TWFE

- When units are treated **at different times**, the single TWFE coefficient becomes a weighted average of all possible 2×2 comparisons — including ones that use **already-treated units as controls** ("forbidden comparisons").
- With dynamic effects, those weights can even go **negative**, so $\hat{\tau}$ can have the wrong sign (Goodman-Bacon 2021; de Chaisemartin & D'Haultfœuille 2020).
- Fix: **heterogeneity-robust estimators** that only compare treated to not-yet-treated units (Callaway & Sant'Anna 2021). See the **appendix**.

Synthetic Control

SC: we predict the counterfactual — as a weighted average

- DiD shifts the *whole* control group to the treated unit's level. But what if **no single control** looks like the treated unit?
- **Synthetic control** builds a bespoke control: a **weighted average of donor units** that tracks the treated unit's pre-treatment path.



Setup and notation

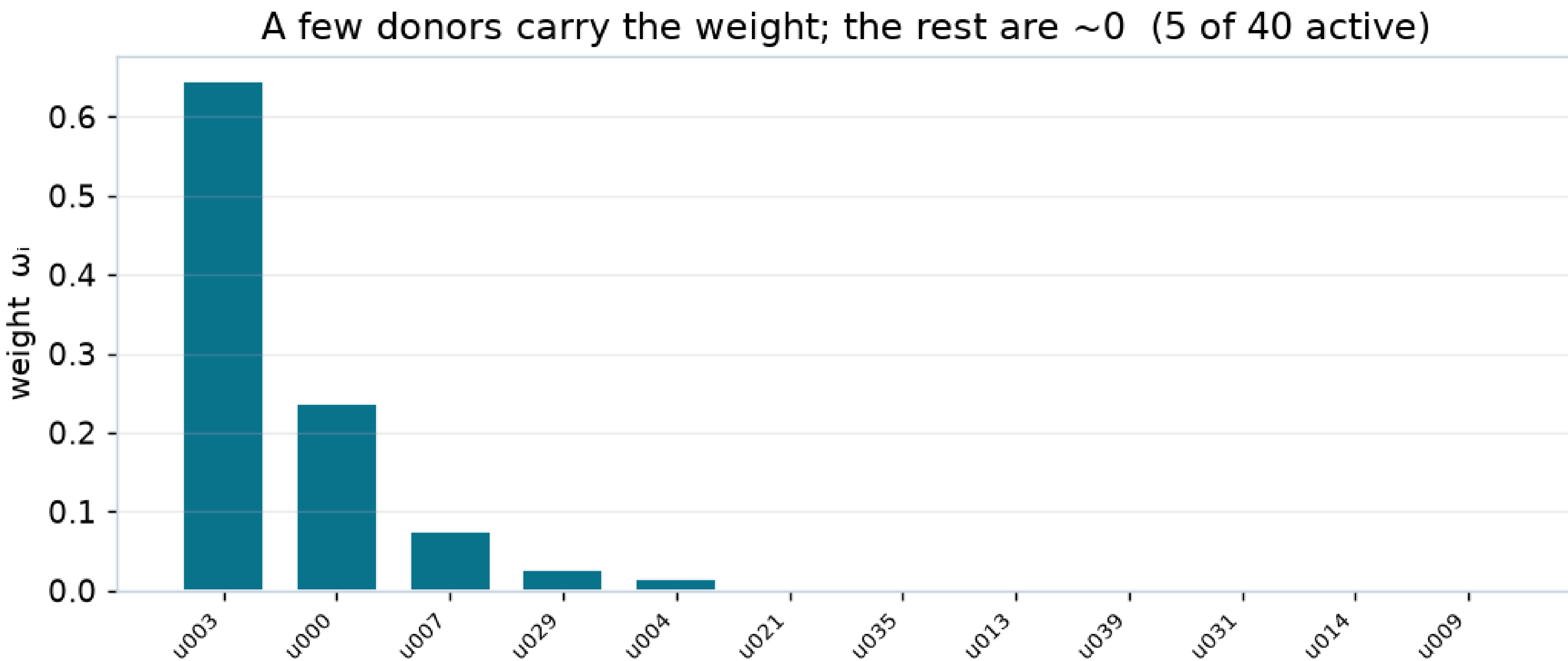
- Let unit $i = 0$ be the **treated** unit; units $i = 1, \ldots, J$ are the **donor pool** of untreated controls.
- We predict the treated counterfactual as an intercept plus a weighted sum of donors:

$$\hat{Y}_{0t}(0) = \mu + \sum_{j=1}^{J} \omega_j Y_{jt}$$

- Choose (μ, ω) so the synthetic unit **matches the treated unit's pre-treatment outcomes** Y_{0t} for $t < 0$.
- The estimated effect is the post-period gap: $\hat{\tau}_t = Y_{0t} - \hat{Y}_{0t}(0)$.

The big idea: a data-driven control group

- Rather than assume a parallel trend, SC **lets the data pick** which controls, and in what proportions, reproduce the treated unit's history.
- Good pre-period fit is **checkable** — you can see whether the synthetic unit tracks the treated one before treatment.
- The weights are usually **sparse**: a handful of donors do the work.


A few donors carry the weight; the rest are ~0 (5 of 40 active)
weight ω_i
0.0
0.1
0.2
0.3
0.4
0.5
0.6
u003
u000
u007
u029
u004
u021
u035
u013
u039
u031
u014
u009

Abadie–Diamond–Hainmueller (2010): the classic SC

- The original SC restricts the weights to a **simplex**: non-negative and summing to one, with no intercept.

$$\min_{\omega} \sum_{t<0} \left(Y_{0t} - \textstyle\sum_j \omega_j Y_{jt} \right)^2 \quad \text{s.t.} \quad \omega_j \geq 0,\ \textstyle\sum_j \omega_j = 1$$

- This keeps the counterfactual **inside the convex hull** of the donors — no extrapolation, interpretable weights.
- Best when the treated unit is *bracketed* by the donor pool. In `panelib`: `sc.sc_model(model_name='adh', ...)`.

Doudchenko–Imbens (2016): a less restrictive SC

- Relax the ADH constraints: allow an **intercept** μ (a level shift, like DiD) and let the weights be estimated with an **elastic-net penalty** instead of the simplex.

$$\min_{\mu,\omega} \sum_{t<0} \Big(Y_{0t} - \mu - \textstyle\sum_j \omega_j Y_{jt}\Big)^2 + \lambda\Big[(1-\alpha)\|\omega\|_2^2 + \alpha\|\omega\|_1\Big]$$

- The penalty (λ, α) is chosen by **cross-validation** on held-out pre-periods, trading off fit against over-fitting the donor weights.
- More flexible, but can extrapolate. In `panelib`: `sc.sc_model(model_name='di', ...)`.

How does inference work for SC?

- With one treated unit there is no cross-sectional sample to form a standard error the usual way. Two common answers:
- **Placebo / permutation** (ADH): re-run SC pretending **each donor** was the treated unit. If the real treated gap is extreme relative to this placebo distribution, it's significant (a Fisher-style exact test).
- **Conformal inference** (Chernozhukov, Wüthrich, Zhu 2021): invert a test that permutes residuals across time blocks — valid with a single treated unit, and the procedure `panelib` uses for **all** the models here, so the comparison is apples-to-apples.

Validating SC and adding covariates

- **Pre-period fit** is the first diagnostic: a synthetic unit that can't track the pre-period won't be trusted post-period. `panelib` reports pre-train and pre-test fit (MAPE) so you can catch over-fitting.
- **Backdating / hold-out**: fit on early pre-periods, check the fit on the held- out late pre-periods.
- **Covariates**: ADH can match on pre-treatment covariates as well as outcomes; DI-style models fold them into the donor design matrix. See Abadie (2021) for a full review.

Synthetic Difference-in-Differences

Motivation: why hedge?

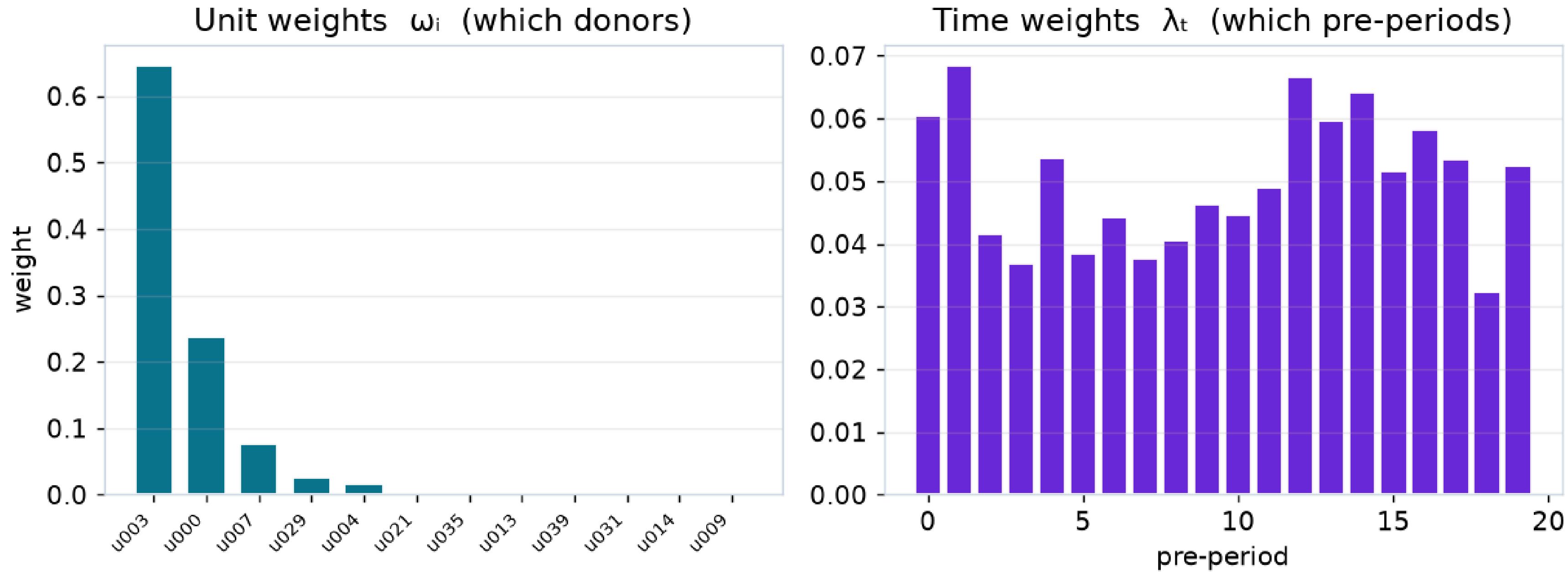
- **DiD** trusts *parallel trends* — one bad assumption and it can break badly.
- **Synthetic control** trusts *pre-period fit* — great when a good donor mix exists, but it can extrapolate and it ignores the level information DiD uses.
- **Synthetic DiD** (Arkhangelsky, Athey, Hirshberg, Imbens, Wager 2021) is a **hedge**: keep DiD's double-differencing, but *weight* the controls (like SC) and *weight* the time periods, so parallel trends only has to hold for the weighted comparison.

SDID: we predict the counterfactual — weighted, then differenced

- SDID runs a **weighted two-way fixed effects** regression:

$$\hat{\tau} = \arg\min_{\tau,\mu,\eta,\gamma} \sum_{i,t} \left(Y_{it} - \mu - \eta_i - \gamma_t - \tau D_{it}\right)^2 \hat{\omega}_i \, \hat{\lambda}_t$$

- Two sets of weights do the hedging:
  - **Unit weights** $\hat{\omega}_i$ — SC-style, pick controls whose *trend* matches the treated unit (an intercept is allowed, so only trends matter).
  - **Time weights** $\hat{\lambda}_t$ — pick pre-periods that look like the post-period, down-weighting irrelevant history.


Unit weights ωᵢ (which donors)
weight
0.0
0.1
0.2
0.3
0.4
0.5
0.6
u003
u000
u007
u029
u004
u021
u035
u013
u039
u031
u014
u009
Time weights λₜ (which pre-periods)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0
5
10
15
20
pre-period

What the two weightings buy you

- **Unit weights** make parallel trends only need to hold **between the treated unit and its synthetic match**, not the whole donor pool — a much weaker ask.
- **Time weights** focus the "before" comparison on the pre-periods most like the post-period, reducing sensitivity to distant, irrelevant history.
- The DiD double-difference is still there, so SDID keeps DiD's robustness to *level* differences that pure SC has to fit away.
- In `panelib`: `sdid.twfe_sdid(...)`, with `sdid.conformal_inference(...)` for the CI — the same conformal procedure as the other models.

When does SDID help?

- SDID **shines when parallel trends is shaky but a good weighted control exists** — it often has the lowest bias *and* keeps close-to-nominal coverage.
- It **reduces to DiD** if the data want uniform weights, and behaves **like SC** if the level shift is unnecessary — so it rarely does much *worse* than either.
- Cost: it needs a reasonable number of pre-periods to estimate the time weights, and inference is a bit more involved. We will see all of this in the worked example.

DiD vs. SC vs. SDID

	DiD	Synthetic Control	Synthetic DiD
Control group	unweighted average of all controls, allowing a level shift	weighted subset matching the treated pre-period	weighted controls **and** a level shift
Key assumption	parallel trends (all controls)	good pre-period fit	parallel trends for the **weighted** match
Data it wants	many units *or* many periods	long pre-period, good donors	both, ideally
Weights	none (uniform)	unit weights ω	unit **and** time weights ω, λ
Inference	OLS / clustered; here, conformal	permutation / conformal	conformal
Main failure mode	non-parallel trends	no donor mix fits; extrapolation	too few pre-periods

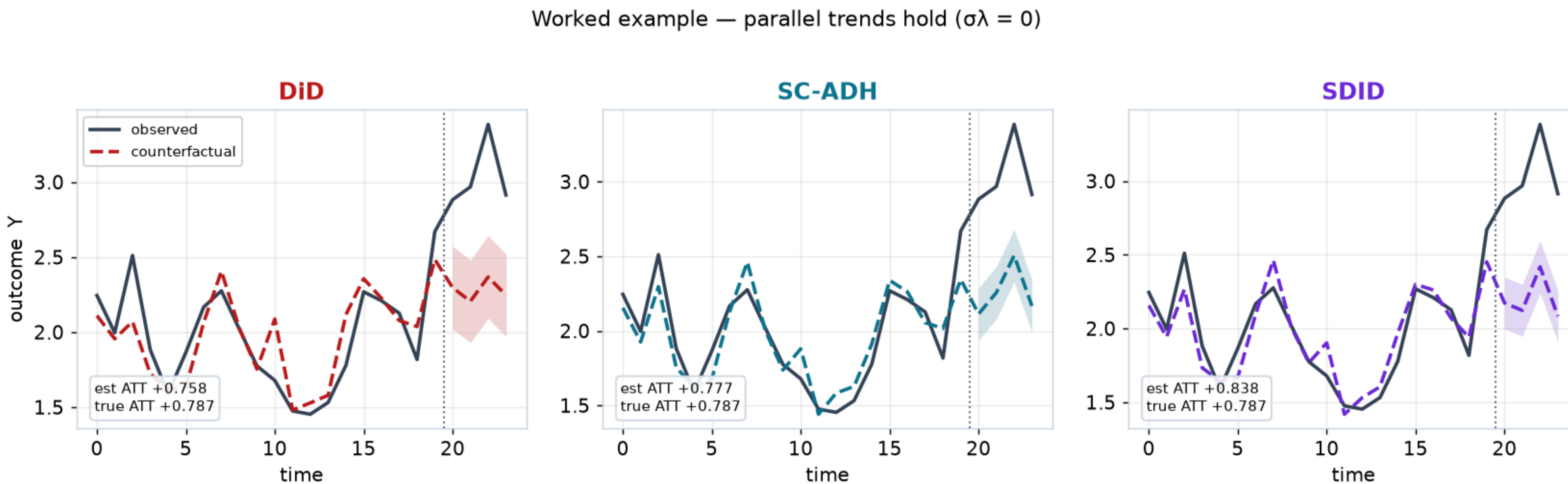
Worked Example

Simulated data, so we know the answer

- We simulate a panel with **one treated unit** and **100 controls**, 20 pre-periods and a short post-period, using `panelib`'s `dgp.simulate_panel`. Because we built it, we know the **true ATT**.
- A shared **AR(1) common trend** ($\rho = 0.8$) drives all units, plus unit effects and noise. Two scenarios:
    1. **Parallel trends hold** ($\sigma_\lambda = 0$): no unit-specific trends.
    2. **Parallel trends violated** ($\sigma_\lambda = 1$): each unit gets its own linear trend, so the treated unit drifts away from the controls.
- We fit **DiD, SC-ADH, SC-DI, and SDID**, all with the *same* conformal inference. Companion notebook with all code here.

Scenario 1: parallel trends hold

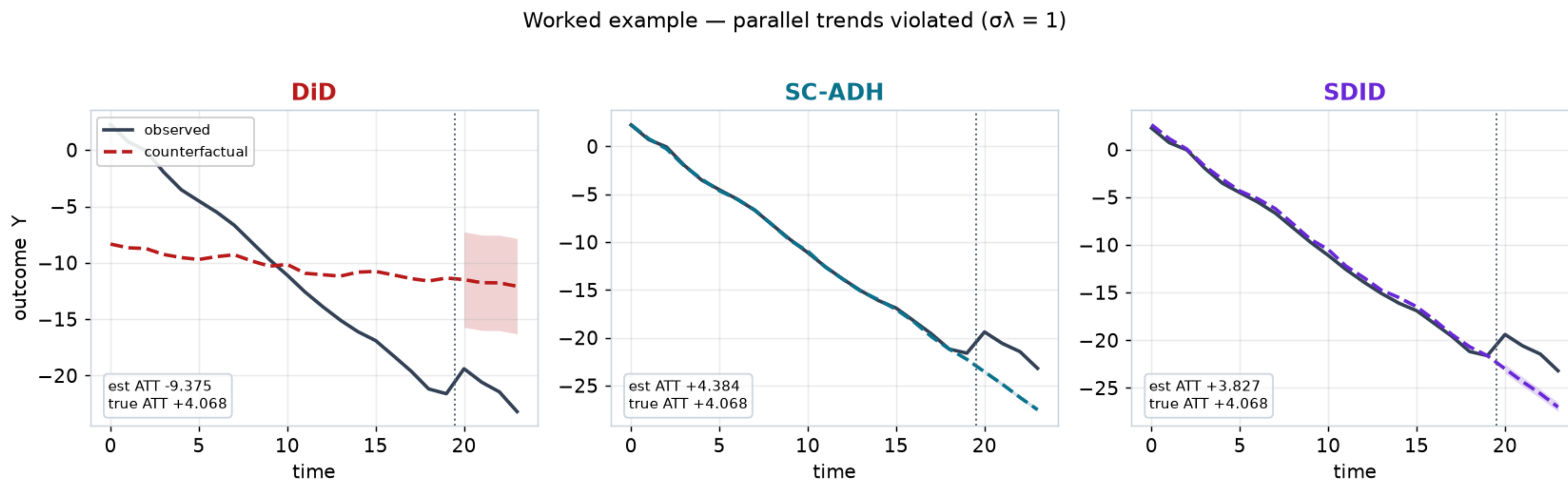
- Observed treated outcome (solid) vs. each model's predicted counterfactual (dashed), with a 95% conformal band in the post-period.



- All three track the treated unit's pre-period and land near the true ATT — when parallel trends holds, every method works.

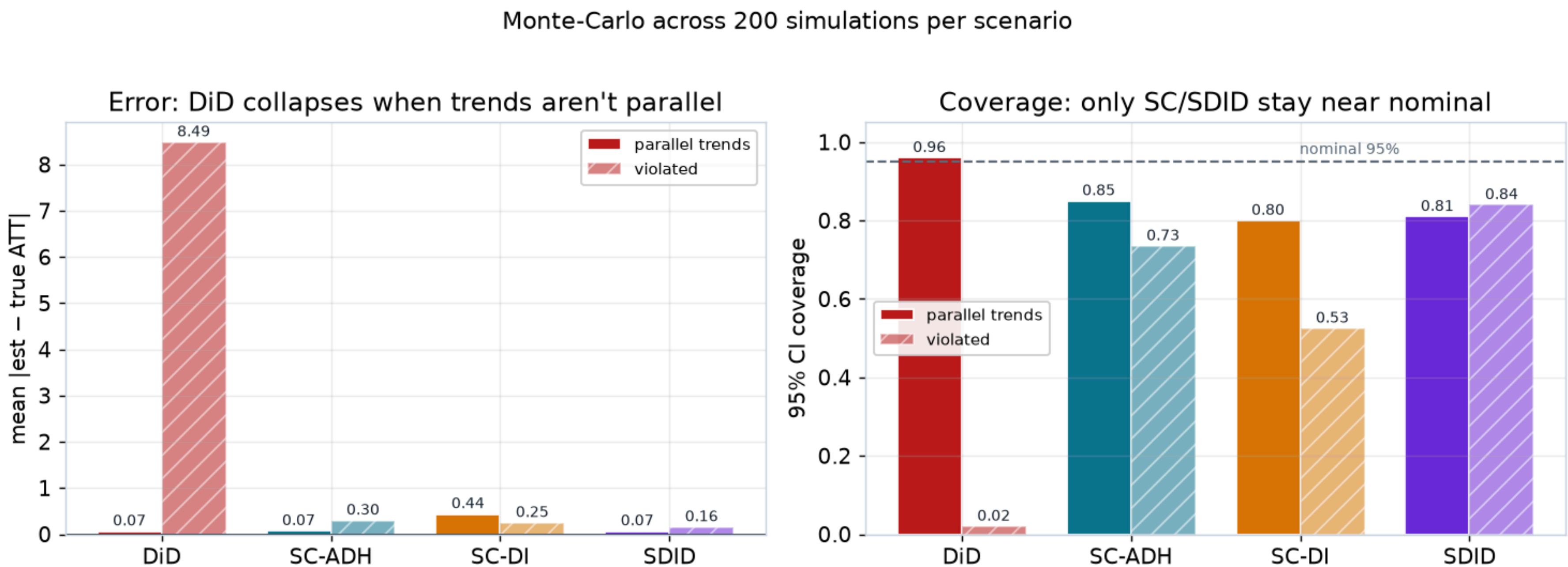
Scenario 2: parallel trends violated

- Now the treated unit has its own downward trend the controls don't share.



- **DiD's counterfactual ignores that trend and misses badly.** SC-ADH and SDID weight the donors to reproduce the treated unit's trajectory, so their counterfactuals bend with it and stay close to the truth.

Error and coverage across 200 simulations per scenario



- **Error (left):** under a violation **DiD misses by ~8.5** (SC/SDID ~0.2). DiD's *signed* bias averages to ~0 — trends run both ways — so absolute error is the honest read.
- **Coverage (right):** DiD **96% → 2%**; SC/SDID stay near nominal (**SDID 84%**), though only **80–85%** even under parallel trends (2 post-periods ⇒ coarse permutation p-value).

Reading the results

- **DiD** is efficient and unbiased *when its one assumption holds* — and catastrophic when it doesn't. Its confidence interval doesn't warn you: it stays tight while pointing at the wrong answer.
- **Synthetic control** is robust to the trend violation because it fits the treated trajectory directly — at the cost of wider, model-dependent inference.
- **SDID** gets the best of both here: low bias under the violation *and* the coverage closest to nominal, because the weighting makes its parallel-trends requirement much weaker.

Conclusion — which method, when?

- Start from the **counterfactual** you can defend, not the estimator you like.
- **DiD**: reach for it when parallel trends is plausible and you have clean timing. Cheap, transparent, and the event study makes the assumption visible.
- **Synthetic control**: when one treated unit (or a few) has a long pre-period and a good donor pool — and you can *show* the pre-period fit.
- **Synthetic DiD**: when you're worried about trends but have the panel to support weighting — often the safest default.
- Whatever you pick: **state the assumption, test what you can, and stress-test what you can't.**

Appendix

Appendix: panel data's advantage over cross-sections

- With a single cross-section, a level difference between treated and control units (η_i) is indistinguishable from a treatment effect.
- Panel data lets us **difference out** the time-invariant η_i by comparing each unit to its own past — the core reason panel methods can beat selection-on-observables.
- The price is a **new** assumption about *time-varying* differences (parallel trends), which is what the rest of this deck is about.

Appendix: staggered adoption and forbidden comparisons

- With a single treatment date, TWFE = the clean 2×2 DiD. With **staggered** timing, Goodman-Bacon (2021) shows $\hat{\tau}^{TWFE}$ is a variance-weighted average of **all** 2×2 comparisons between timing groups.
- Some of those comparisons use **already-treated units as the control group**. If treatment effects grow over time, the already-treated unit's *rising* effect enters with a **negative weight**, biasing $\hat{\tau}$ — possibly to the wrong sign.
- **Heterogeneity-robust estimators** (Callaway & Sant'Anna 2021; Sun & Abraham 2021; de Chaisemartin & D'Haultfœuille 2020) fix this by only ever comparing newly-treated units to **not-yet-treated** ones, then aggregating clean group-time effects $ATT(g, t)$.

Appendix: the shared conformal inference

- All models here share one inference procedure, so their intervals are comparable (`conformal_inf` in `panelib`).
- **Idea** (Chernozhukov, Wüthrich & Zhu 2021): under the null $\tau_t = \theta$, the post-period residuals (observed − counterfactual − θ) should look exchangeable with the pre-period residuals. Permute residuals across time blocks, compute a test statistic, and get a p-value.
- Invert the test over a grid of θ to get the confidence interval; the reported SE is the CI width $/\,(2 \times 1.96)$.
- Because it keys off **pre-period fit**, models that fit the pre-period poorly (DiD under a trend violation) get **wider, honest** intervals — usually.
- A dedicated deck — **Part 10, Conformal Inference** — develops this method in full, with a synthetic-control worked example.

Appendix: estimating the SDID weights

- **Unit weights** $\hat{\omega}$ solve an SC-style problem *with an intercept and a small ridge penalty*, matching control pre-period trends to the treated unit:

$$\min_{\omega_0, \omega \geq 0,\, \mathbf{1}'\omega = 1} \sum_{t<0} \left(\omega_0 + \sum_j \omega_j Y_{jt} - Y_{0t} \right)^2 + \zeta^2 T_{pre} ||\omega||_2^2$$

- **Time weights** $\hat{\lambda}$ solve the analogous problem *across time*, finding the pre-periods whose control outcomes best predict the post-period.
- Plug both into the **weighted TWFE** regression; the coefficient on D_{it} is the SDID ATT. `panelib`: `sdid.twfe_sdid` returns `omega_weights`, `lambda_weights`, and the counterfactual.

Appendix: inference with many treated units

- This deck's worked example has **one treated unit** — that is where conformal inference earns its keep. With **many treated units**, the workhorse is the TWFE regression with **cluster-robust SEs at the unit level**.
- Clustering matters because outcomes are **serially correlated within units** — ignoring it understates DiD standard errors badly (Bertrand, Duflo & Mullainathan 2004).
- **Few clusters** (a handful of regions or cohorts) breaks the cluster asymptotics: use the **wild cluster bootstrap** (Cameron, Gelbach & Miller 2008) or randomization inference.
- Rule of thumb: many treated units → cluster; one or a few → permutation / conformal (this deck); in between → wild bootstrap plus sensitivity checks.
- The companion notebook runs a clustered-SE DiD next to the conformal interval.

Appendix: panel methods in industry

- **Geo experiments / marketing measurement.** Regions are the units; a few get the campaign. That is exactly the SC/SDID setting — small donor pools, aggregate outcomes, one intervention date.
- **Staggered feature launches** are the staggered-adoption trap from the DiD appendix: a single TWFE across launch waves mixes clean and forbidden comparisons.
- **Interference.** Treatment that leaks across units (marketplace equilibria, network effects, border-crossing shoppers) contaminates the donor pool — pick controls the launch cannot touch, or aggregate until it can't.
- The estimand stays the **ATT of the units you launched on** — extrapolating to a full rollout is an extra assumption, not a free lunch.

Literature Review of Related Papers

- **DiD — foundations & modern advances**
  - Roth, Sant'Anna, Bilinski, Poe (2023). "What's Trending in DiD? A Synthesis of the Recent Econometrics Literature." *J. Econometrics.* link
  - Goodman-Bacon (2021). "DiD with Variation in Treatment Timing." *J. Econometrics.* link
  - Callaway & Sant'Anna (2021). "DiD with Multiple Time Periods." *J. Econometrics.* link
  - de Chaisemartin & D'Haultfœuille (2020). "Two-Way FE Estimators with Heterogeneous Treatment Effects." *AER.* link
  - Sun & Abraham (2021). "Estimating Dynamic Treatment Effects in Event Studies." *J. Econometrics.* link
  - Borusyak, Jaravel, Spiess (2024). "Revisiting Event-Study Designs: Robust and Efficient Estimation." *REStud.* link

Literature Review (continued)

- **DiD — inference, covariates, and pre-tests**
  - Bertrand, Duflo, Mullainathan (2004). "How Much Should We Trust Differences-in-Differences Estimates?" *QJE.* link
  - Cameron, Gelbach, Miller (2008). "Bootstrap-Based Improvements for Inference with Clustered Errors." *REStat.* link
  - Sant'Anna & Zhao (2020). "Doubly Robust DiD Estimators." *J. Econometrics.* link
  - Roth (2022). "Pre-test with Caution: Event-Study Estimates After Testing for Parallel Trends." *AER: Insights.* link
  - Rambachan & Roth (2023). "A More Credible Approach to Parallel Trends." *REStud.* link

Literature Review (continued)

- **Synthetic control**
    - Abadie & Gardeazabal (2003). "The Economic Costs of Conflict: A Case Study of the Basque Country." *AER.* link
    - Abadie, Diamond, Hainmueller (2010). "Synthetic Control Methods for Comparative Case Studies." *JASA.* link
    - Doudchenko & Imbens (2016). "Balancing, Regression, DiD and Synthetic Control Methods." *NBER w22791.* link
    - Ferman & Pinto (2021). "Synthetic Controls with Imperfect Pretreatment Fit." *Quantitative Economics.* link
- **Synthetic DiD & shared inference**
    - Arkhangelsky, Athey, Hirshberg, Imbens, Wager (2021). "Synthetic Difference-in-Differences." *AER.* link
    - Chernozhukov, Wüthrich, Zhu (2021). "Conformal Inference for Counterfactual and Synthetic Controls." *JASA.* (See **Part 10**.) link

- **Companion code:** `panelib.py` in statanomics.